

D783

DevOps Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A retailer's developers hand finished code to a separate operations team every six weeks. Each handoff contains hundreds of changes, and the work sits in a queue for days before operations can install it. The operations team recently moved to a new office across town. Which DevOps principle most directly targets this problem?

- A. Move the operations team back into the developers' building.
- B. Hire more operations staff so each handoff is installed faster.
- C. The Third Way: build a culture of experimentation and learning.
- D. The First Way: speed the flow of small batches from development to operations.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A team of eight developers, all working in Java on a well-known hosted Git service, keeps features on separate branches for about three weeks. Merging at the end regularly takes two days of conflict fixing. Which practice best fixes this?

- A. Move the code to a different Git hosting service.
- B. Merge small changes into the main branch at least daily, with an automated build and test on each merge.
- C. Keep branches open until each feature is fully finished, then merge once.
- D. Add a manual QA pass on each branch before the three-week merge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A team says it practices CI because a server builds the main branch every night. Developers merge their branches into main about every two weeks. What is missing?

- A. Frequent integration: small changes merged to main at least daily.
- B. Automatic deployment of every build to production.
- C. Nothing; any automated build server means the team practices CI.
- D. A code-review approval before the nightly build runs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A payments team's pipeline builds, tests, and packages every commit and deploys it to a staging environment that mirrors production. A release manager clicks an approval button to send a release to production, which happens several times a week. The team runs 14 services across two cloud regions. Which practice does this pipeline show?

- A. Continuous integration only.
- B. Continuous deployment.
- C. Continuous delivery.
- D. Not continuous at all, because any manual step breaks the pipeline.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team wants every change that passes all automated tests to reach production with no human approval. Which practice is that, and what does it remove?

- A. Continuous integration; it removes long-lived branches.
- B. Continuous delivery; it removes the manual release approval.
- C. Continuous deployment; it adds a change board review of each change.
- D. Continuous deployment; it removes the manual release approval.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Engineers build new servers by hand from a 40-page wiki page last edited by a contractor. Two servers that should be identical behave differently in testing. Which approach best prevents this?

- A. Define the servers in version-controlled code that an automation tool applies the same way every time.
- B. Take nightly backups of each server.
- C. Assign one senior engineer to build every server by hand.
- D. Have a staff engineer rewrite and sign off on the wiki page.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

An engineer runs an idempotent infrastructure-as-code configuration against a server that already matches it exactly. What should happen?

- A.** The server is destroyed and rebuilt every run.
- B.** The run fails because the resources already exist.
- C.** Nothing changes; the tool reports the server is already in the desired state.
- D.** A second copy of every resource is created.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A web tier of 20 servers has been patched in place by several admins over two years. A new release fails on 3 of them because of undocumented changes. The servers run on hardware from a large, reputable vendor. What best prevents this from happening again?

- A.** Stop applying all patches so nothing changes.
- B.** Build servers from a versioned image and replace them instead of changing them in place.
- C.** Give admins a longer checklist for in-place patching.
- D.** Switch to a different hardware vendor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In one quarter a team deployed to production 200 times. Thirty of those deployments caused a failure that needed a rollback or hotfix. The team also opened 450 tickets and has 9 engineers. What is its change failure rate?

- A.** 6.7%
- B.** 85%
- C.** 15%
- D.** 3.3 failures per engineer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A team's change failure rate last month was 8%, and it had 12 failed deployments. How many deployments did it make?

- A.** 96
- B.** 150
- C.** 138
- D.** 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A team finds most defects during a two-week manual test phase just before each release, and fixes then take days. The test lab is in a separate building. Leadership wants defects to be cheaper to fix. What should the team do?

- A.** Drop pre-release testing and rely on production monitoring.
- B.** Extend the manual test phase to four weeks.
- C.** Move the test lab into the developers' building.
- D.** Run automated unit and static checks on every commit so defects surface within minutes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A 90-minute outage began when an engineer ran a database script against production instead of test. The engineer has worked there six years and is well liked. Both environments use the same login screen and prompt. What should the postmortem focus on?

- A.** The system conditions that made the mistake easy, such as look-alike environments, and fixes for them.
- B.** Formally disciplining the engineer so others are more careful.
- C.** Skipping a review because of the engineer's long, strong record.
- D.** Banning all scripts forever so this can never happen again.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A retailer's developers hand finished code to a separate operations team every six weeks. Each handoff contains hundreds of changes, and the work sits in a queue for days before operations can install it. ~~The operations team recently moved to a new office across town.~~ Which DevOps principle most directly targets this problem?

Target word: which principle fits large, queued handoffs **Decides:** Big batches waiting between teams is a flow problem.

A	Camouflage	Answers the office move; location does not shrink six-week batches.
B	Adjacent category	Speeds one silo but keeps the large batches and the handoff.
C	Confusion set	A real DevOps principle, but it addresses learning, not queued handoffs.
D	Correct	Flow is about shrinking batch size and removing queues across the whole value stream.

2. Answer B Concept 2

A team of eight developers, ~~all working in Java on a well-known hosted Git service,~~ keeps features on separate branches for about three weeks. Merging at the end regularly takes two days of conflict fixing. Which practice best fixes this?

Target word: fix for painful late merges **Decides:** Long-lived branches merged rarely.

A	Camouflage	The tool is not the problem; merge frequency is.
B	Correct	That is continuous integration; small, frequent merges keep conflicts tiny.
C	Inverted operation	Longer branches make the conflicts worse.
D	Dropped element	Adds testing but drops the frequent merge that CI requires.

3. Answer A Concept 2 · transfer

A team says it practices CI because a server builds the main branch every night. Developers merge their branches into main about every two weeks. What is missing?

Target word: what CI requires beyond a build server **Decides:** Integration frequency is the core of CI.

A	Correct	Without frequent merging, the nightly build only tests stale code.
B	Adjacent category	That is continuous deployment, a later stage.
C	Absolute language	A build server is a tool; CI is the habit of integrating often.
D	Confusion set	Reviews are useful but do not fix the two-week integration gap.

4. Answer C Concept 3

A payments team's pipeline builds, tests, and packages every commit and deploys it to a staging environment that mirrors production. A release manager clicks an approval button to send a release to production, which happens several times a week. ~~The team runs 14 services across two cloud regions.~~ Which practice does this pipeline show?

Target word: name the practice **Decides:** Every change is releasable, but a human approves production.

A	Dropped element	The pipeline goes beyond build and test to releasable, staged packages.
B	Confusion set	Continuous deployment has no manual approval before production.
C	Correct	Always releasable, with a manual decision to release.
D	Absolute language	A manual release gate is exactly what defines continuous delivery.

5. Answer D Concept 3 • transfer

A team wants every change that passes all automated tests to reach production with no human approval. Which practice is that, and what does it remove?

Target word: name the practice from the outcome **Decides:** No human approval before production.

A	Adjacent category	CI covers merging and testing, not releasing to production.
B	Confusion set	Continuous delivery keeps the human release decision.
C	Inverted operation	Adds a human gate instead of removing it.
D	Correct	Passing changes go straight to production automatically.

6. Answer A Concept 4

Engineers build new servers by hand from a 40-page wiki page ~~last edited by a contractor.~~ Two servers that should be identical behave differently in testing. Which approach best prevents this?

Target word: prevent differences between built environments **Decides:** Manual builds from a document are not repeatable.

A	Correct	Infrastructure as code makes builds repeatable and reviewable.
B	Adjacent category	Backups help recovery, not consistent builds.
C	Dropped element	Still manual, so still error-prone and slow.
D	Camouflage	Answers who edited the page; builds are still manual.

7. Answer C Concept 4 • transfer

An engineer runs an idempotent infrastructure-as-code configuration against a server that already matches it exactly. What should happen?

Target word: behavior of an idempotent run **Decides:** Idempotent means repeat runs produce the same state.

A	Adjacent category	That describes immutable replacement, not an idempotent apply.
B	Confusion set	A declarative tool compares to desired state; existing resources are expected.
C	Correct	Idempotence means applying it again has no further effect.
D	Inverted operation	Duplicating resources is the opposite of idempotence.

8. Answer B Concept 5

A web tier of 20 servers has been patched in place by several admins over two years. A new release fails on 3 of them because of undocumented changes. ~~The servers run on hardware from a large, reputable vendor.~~ What best prevents this from happening again?

Target word: prevent configuration drift **Decides:** In-place changes over time caused the drift.

A	Absolute language	Leaves servers insecure; the fix is how changes are made, not whether.
B	Correct	Immutable infrastructure means no server can drift from its definition.
C	Dropped element	Keeps in-place changes, the source of drift.
D	Camouflage	Hardware did not cause the undocumented changes.

9. Answer C Concept 6

In one quarter a team deployed to production 200 times. Thirty of those deployments caused a failure that needed a rollback or hotfix. ~~The team also opened 450 tickets and has 9 engineers.~~ What is its change failure rate?

Target word: change failure rate **Decides:** Failed deployments divided by total deployments.

A	Wrong stem number	30 / 450 uses tickets as the base.
B	Inverted operation	That is the share of deployments that succeeded.
C	Correct	$30 / 200 = 0.15$.
D	Camouflage	Team size does not decide change failure rate.

10. Answer B Concept 6 · transfer

A team's change failure rate last month was 8%, and it had 12 failed deployments. How many deployments did it make?

Target word: solve for total deployments **Decides:** Total = failures / rate.

A	Inverted operation	Multiplies 12×8 instead of dividing.
B	Correct	$12 / 0.08 = 150$.
C	Dropped element	That is only the successful deployments; the failed 12 are dropped.
D	Wrong stem number	The failed deployments, not the total.

11. Answer D Concept 7

A team finds most defects during a two-week manual test phase just before each release, and fixes then take days. ~~The test lab is in a separate building.~~ Leadership wants defects to be cheaper to fix. What should the team do?

Target word: make defects cheaper to fix **Decides:** Defects are found late.

A	Adjacent category	Monitoring complements testing but finds defects later, after customers are hit.
B	Inverted operation	Keeps testing at the end and delays feedback further.
C	Camouflage	Location does not change when defects are found.
D	Correct	Shifting testing left catches defects when they are cheapest to fix.

12. Answer A Concept 8

A 90-minute outage began when an engineer ran a database script against production instead of test. ~~The engineer has worked there six years and is well liked.~~ Both environments use the same login screen and prompt. What should the postmortem focus on?

Target word: postmortem focus **Decides:** Identical environments made the error easy to make.

A	Correct	Blameless reviews fix the conditions so anyone would be less likely to repeat the error.
B	Inverted operation	Blame hides information and leaves the look-alike environments in place.
C	Camouflage	Tenure does not decide whether the system needs fixing.
D	Absolute language	Overreacts and blocks needed work; targeted safeguards fit better.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The First Way: flow	DevOps first optimizes fast, smooth flow of small batches from development to operations across the whole system, not one team's output.	Thinking DevOps means making each silo faster on its own.	Large handoffs, queues, and waiting between teams.
2	Continuous integration	Developers merge small changes into a shared mainline at least daily, and every merge triggers an automated build and test.	Believing a build server alone equals CI, even with long-lived branches.	Painful merges after weeks on separate branches.
3	Continuous delivery vs. continuous deployment	Continuous delivery keeps every change releasable with a human deciding when to release; continuous deployment releases every passing change automatically.	Using the two terms as synonyms.	Whether a manual approval sits before production.
4	Infrastructure as code	Environments are defined in version-controlled, idempotent code so the same definition always builds the same result.	Treating better written instructions for manual builds as IaC.	Hand-built servers that should be identical but are not.
5	Configuration drift and immutable infrastructure	Replace servers from a known image instead of changing them in place, so drift cannot build up.	Fixing drift with more careful in-place changes.	Servers patched by hand over years with undocumented differences.
6	DORA metrics	Deployment frequency and lead time measure throughput; change failure rate (failed deployments / total deployments) and time to restore measure stability.	Dividing failures by the wrong base, such as tickets or engineers.	Counts of deployments and failed deployments in one stem.
7	Shift-left testing and fast feedback	Automated tests run early and on every change, because defects found minutes after they are written are cheapest to fix.	Adding more manual testing at the end of the cycle.	Defects found late in a long test phase.
8	Blameless culture	Postmortems look for the system conditions that allowed a failure and fix those, not for a person to punish.	Assuming discipline prevents repeat errors.	An outage traced to one person's action.

Course scope (public description): The DevOps Foundations course will provide students with a comprehensive introduction to the core principles and practices of DevOps, focusing on infrastructure, configuration, software engineering/operations, and the cultural shifts necessary for successful implementation. Students will understand the history, principles, and business benefits of DevOps, along with practical skills in utilizing basic toolchains and implementing automation and continuous processes throughout the DevOps lifecycle.